ETALE-LOCALLY LIFTABLE MORPHISM DATA BETWEEN FIRST ORDER THICKENINGS FORMS A SHEAF ON THE SMALL ETALE SITE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3790-3813 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3815-3824, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (spaces, more general; both sides complete). Gate0 note: label lemma-sheaf is generic but the claim_key is source-qualified; occupancy confirmed FREE.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Descent on Spaces, lemma fpqc-universal-effective-epimorphisms; definition of first order thickening and induced conormal map (in-source).

Source reference. Original lemma 04P2: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3824–3824. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $X \subset X'$ and $Y \subset Y'$ be first order thickenings over B. Assume given a morphism $a : X \to Y$ and a map $A : a^*\mathcal{C}_{Y/Y'} \to \mathcal{C}_{X/X'}$ of $\mathcal{O}_X$-modules. For an object U' of $(X')_{spaces,étale}$ with $U = X \times_{X'} U'$ consider morphisms $a' : U' \to Y'$ such that*

(1) *a' is a morphism over B,*
(2) *$a'|_U = a|_U$, and*
(3) *the induced map $a^*\mathcal{C}_{Y/Y'}|_U \to \mathcal{C}_{X/X'}|_U$ is the restriction of A to U.*

Then the rule

$$U' \mapsto \{a' : U' \to Y' \text{ such that (1), (2), (3) hold.}\} \tag{0.1.1}$$

defines a sheaf of sets on $(X')_{spaces,étale}$.

Proof. Denote $\mathcal{F}$ the rule of the lemma. The restriction mapping $\mathcal{F}(U') \to \mathcal{F}(V')$ for $V' \subset U' \subset X'$ of $\mathcal{F}$ is really the restriction map $a' \mapsto a'|_{V'}$. With this definition in place it is clear that $\mathcal{F}$ is a sheaf since morphisms of algebraic spaces satisfy étale descent, see Descent on Spaces, Lemma 04P2. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.